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CS 455 Computer Networks
Lab 9
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I. My MAC Address Decoded

The address is: 02:0f:36:a3:12:0d. The search in the IEEE registry yielded nothing. This is because of the fact that MacOS locally administers and randomizes Private WiFi addressing rather than assigning a physical network card.

First Byte in Hex: 02.
In binary: 00000010
I/G Bit: Is a 0, therefore it is a unicast address
U/L Bit: Is a 1, meaning that it is a locally administered address rather than a universal administered MAC address

II. The Phone Comparison

I was able to find the WiFi addresses to my phone and tablet however both run on the “Clarkson Guest” network because I was too lazy to download the enduoram dependencies when I first came to Clarkson. How ever the breakdown was the exact same as my tablet is part of the Apple Ecosystem. The IEEE registry yielded the same results.

My Tablet Address was F6:06:C3:BF:89:86 a
First Byte in Hex: F6.
In binary: 11110110
I/G Bit: Is a 0, therefore it is a unicast address
U/L Bit: Is a 1, meaning that it is a locally administered address rather than a universal administered MAC address

III. One Frame Taken Apart

The Source Address was: 02:0f:36:a3:12:0d
The Destination Address was:10:0e:7e:ac:39:80
Type: IPv4

IV. Address Order

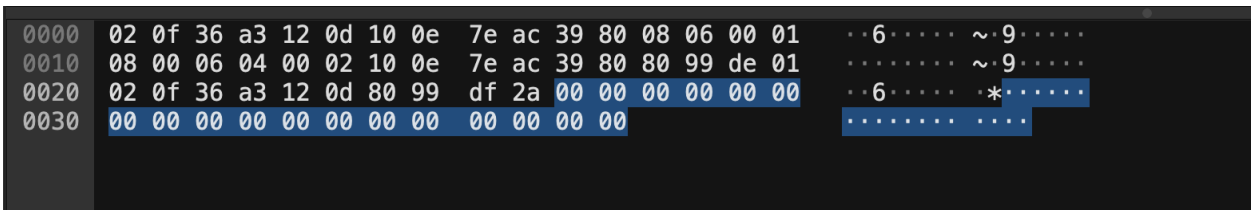
The switch only cares about the destination address and using the forwarding table to move the packet to the next switch. If the source address was listed first this is a further buffer that the switch has to move through to read the header. Ultimately slowing the transmission down. Whereas this does not happen when the Destination address is listed first.

V. The Missing Four Bytes

The Frame is 98 Bytes. The name of the missing field that is 4 bytes long is the FCS/CRC field. Wireshark excludes the frame check sequence from the packet. Thus the arithmetic for the packet is:

Destination MAC address: 6 bytes
+
Source MAC address: 6 bytes
+
Ether Type Length: 2 bytes
+
Payload: 84 bytes
+
Frame Check Sequence: 4 bytes
==========
Total Frame Size on the Wire: 102 bytes

VI. The 60-Byte Frame



This frame exists because IEEE standard say that the frame length mist be 60 bytes. Clearly here only 42 of the 60 are being used. Thus the trailing 0's are to make the frame reach the minimum length.

VII. Broadcast and Multicast

The protocol inside is ARP
01:00:5e:00:00:fb (IPv4)
33:33:00:00:00:fb (IPv6).

Hex = 01 ends in 1 in binary and so does hex = 33 so both I/G bits are set to 1.

Non ARP I/G bit:

02:0f:36:a3:12:0d
Hex 02 ends in a 0 in binary so the I/G bit is set to 0

VIII. The ARP Exchange

1)
Destination: ff:ff:ff:ff:ff:ff

Why? An ARP request uses the Ethernet broadcast address so that every machine on the local network segment receives the frame, allowing the host that owns the targeted IP address to hear the request and respond.

English Question: Who has the IP address specified in the request (e.g., the default gateway's IP), and what is your MAC address so I can send traffic to you?

2)

Destination: 02:0f:36:a3:12:0d

Why they are not the same: The ARP request was a broadcast because the sender did not know who held the requested IP address. However, the ARP reply is sent as a unicast frame directly back to the specific requester, because the replying host already learned the requester's MAC address from the incoming request frame. There is no need to flood the network with a broadcast reply when the exact destination is known.

3)
Sender Hardware Address: 02:0f:36:a3:12:0d

- Role: Identifies the MAC address of the requesting machine aka. my MAC so the responding host knows where to send the ARP reply frame.

Sender Protocol Address: 128.153.223.42

- Role: Identifies the IPv4 address of the requesting machine so the target knows who initiated the request and can cache the requester's IP-to-MAC mapping.

Target Protocol Address: 128.153.223.1

- Role: Identifies the IPv4 address being queried so that only the machine owning this IP address responds.

Target Hardware Address: 00:00:00:00:00:00

- Role: Represents the unknown MAC address being looked up.

- Why it is all zeros: The requesting host does not yet know the hardware address of the target host. The field is filled with zeros as a placeholder until the target machine responds with its actual MAC address.

IX.Who Learned What

4)

The default gateway (128.153.223.1) also learned my IP-to-MAC mapping.

Which frame: It learned this from Frame 1 (the incoming ARP Request).

Explanation: To send a reply, the gateway needed to know my MAC address. Because my MAC included its own Sender Hardware Address (SHA) and Sender Protocol Address (SPA) inside the request header, the gateway was able to cache my address mapping without needing to send a separate ARP request back to me.

ARP Cache Before And After:

Before:

128.153.222.1 (128.153.222.1) deleted

128.153.222.11 (128.153.222.11) deleted

128.153.222.70 (128.153.222.70) deleted

128.153.222.178 (128.153.222.178) deleted

128.153.223.1 (128.153.223.1) deleted

224.0.0.251 (224.0.0.251) deleted

After: ? (128.153.223.1) at b2:5c:bc:41:1e:37 on en0 ifscope [ethernet]

X. The Address That Is Not There

There are 4 requests within the capture. Spaced approximately a second apart, and nothing came back. The OS ARP process times out after receiving the fourth no reply, It drops the outgoing ICMP packets and throws an internal error. The no reply proves that there is no active host on that IP address at my local Layer 2.

XI. The Gateway Experiment

1)

Field	Remote Ping Frame 32	Local Subnet Ping Frame 205
Destination IP Address	1.1.1.1	128.153.223.1
Destination MAC address	10:0e:7e:ac:39:80	b2:5c:bc:41:1e:37
Whose MAC Address is it	Juniper Networks	My Default Gateway

2)
It belongs to my default Gateway

3)
Even though 1.1.1.1 is thousands of miles away, Ethernet frames operate only on the local link layer and cannot deliver frames directly across network boundaries. My local machine addresses the frame to the local router's MAC address (10:0e:7e:ac:39:80) so the router can receive the physical Ethernet frame. At the first router, these six destination MAC bytes will be stripped off, the IP packet inside will be inspected, and a brand-new Ethernet frame will be constructed with new link-layer addresses to forward the packet toward its next hop.

4)
What Changed:

The Destination IP changed from 1.1.1.1 to 128.153.223.1. The Destination MAC changed from the Juniper gateway MAC (10:0e:7e:ac:39:80) to the specific destination host's MAC (b2:5c:bc:41:1e:37).

What Stayed the Same:

The Source IP (128.153.223.42), Source MAC (02:0f:36:a3:12:0d), EtherType (IPv4 0x0800), and overall frame structure/length remained identical.

XII. Frames 1 and 3

Frame 1 Length: 60 Bytes
Frame 3 Length: 64 Bytes
This is a difference of 4 bytes. This is because Frame 3 has a VLAN tag that takes up exactly 4 bytes of the Ethernet Header.

XIII. The Tag Decoded

TPID: 0x8100
Priority: 0
DEI Bit: 0
VLAN ID: 20

The VLAN ID is the one doing the work of isolating the layer 2 traffic. It identifies which virtual network segment the frame belongs to so switches can forward it only to ports assigned to VLAN 20, keeping traffic logically separated across shared trunk links. The rest of values that were identified in the tag are optional.

XVI. Two VLANs, One Cable

In frame 1 the value is 0x0806. The switch can identify this because of the difference in TPID's. When the switch sees a value that is not 0x8100 it can identify that the 802.1Q header is not present and can parse the rest of the payload. Whereas when it sees the 0x8100 TPID it knows the next 4 bytes are the Tag.
No, the machine that sent Frame 3 will not receive Frame 4. Frame 3 is tagged with VLAN ID 20, while Frame 4 is tagged with VLAN ID 30. Even though both are sent as broadcast frames over the same physical trunk cable, switches enforce VLAN logical isolation, restricting broadcast domain delivery strictly to ports assigned to the same VLAN ID

XV. 1518 Against 1522

When an Ethernet frame travels on the physical wire, it contains a 4-byte Frame Check Sequence (FCS) at the very end. Upon receiving the frame, the Network Interface Card (NIC) calculates the CRC to verify frame integrity. Once verified, the NIC hardware strips off those 4 trailing FCS bytes before passing the packet up to the operating system driver and Wireshark.

XVI. Specifications

1. "The type field of the Ethernet frame must contain the value hexadecimal 0800." In the Wireshark capture from Section 2, Wireshark lists the Type field in the Ethernet II header as 0x0800 (IPv4), matching the exact hex value specified in RFC 894
2. "The minimum length of the data field of a packet sent over an Ethernet is 46 octets. If necessary, the data field should be padded (with octets of zero) to meet the Ethernet minimum frame size. This padding is not part of the IP packet and is not included in the total

length field of the IP header."A 60-byte frame on the wire consists of 14 bytes of header and 46 bytes of data. If the IP payload is under 46 bytes, zeros are padded to the end to bring the length up to the 46 octets.

3. "The minimum length of the data field of a packet sent over an Ethernet is 1500 octets, thus the maximum length of an IP datagram sent over an Ethernet is 1500 octets."The sentence incorrectly states "The minimum length of the data field..." when it meant to say "The maximum length of the data field…". The maximum size of an IP datagram payload that fits into a single standard Ethernet frame is 1500 bytes. There is an obvious error within the specification, however the intent is still good enough to justify the value of the specification
4. "Merge_flag := false If the pair <protocol type, sender protocol address> is already in my translation table, update the sender hardware address field of the entry with the new information in the packet and set Merge_flag to true. ?Am I the target protocol address? Yes: If Merge_flag is false, add the triplet <protocol type, sender protocol address, sender hardware address> to the translation table.” The receiver updates or inserts the sender's <protocol, hardware> mapping into its ARP cache before evaluating whether it is the target machine and before looking at the opcode (REQUEST vs REPLY). This allows the responding machine to automatically learn the requester's MAC address without having to send a separate ARP request.
5. "Periodic broadcasting is definitely not desired. Imagine 100 workstations on a single Ethernet, each broadcasting address resolution information once per 10 minutes (as one possible set of parameters). This is one packet every 6 seconds. This is almost reasonable, but what use is it? ... The protocol described in this paper distributes information as it is needed, and only once (probably) per boot of a machine.” In an active capture, a machine sends multiple IP packets per second. Without an ARP cache, every single outgoing IP packet would require a preceding ARP broadcast request/reply exchange to resolve the next-hop MAC address, generating tens to hundreds of extra ARP frames every minute for basic network communications. With the cache this is not needed, and thus increases the speed of the protocol.

XVII. Reflection

A student like myself and many others is probably stretched financially thin, especially with the rising general cost of education itself. If a standard or specification is free, it gives almost no excuse that students attempting to become professionals wouldn’t be able to use it. If there is a cost associated, a student my be inclined to wait until an employer pays for the specification or just may never learn it. The industry looses out on possible professionals that may be able to contribute to or at least at a minimum be really good at reaching compliance of the specification, if the student decides not to purchase the specification because of the cost.